

Qualifier Question

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1 EXISTING RESOURCES

1.1 Qualifier Question

You have expressed interest in creating an educational tool to teach the concepts of recursion.

First, find two online resources that already exist designed to teach recursion and identify the following:

- Give a high level description of the course and its creator or distributor.
- What role is this resource trying to fill, and who is the audience? Is it teaching recursion from the beginning? Providing a refresher?
- Using research, identify and explain the educationally strong points.
- Also based upon your research, suggest changes/weak points to improve the educational value of the tool.

1.2 A Structured Approach to Teaching Recursion Using Cargo-Bot

Elynn Lee et al. (2014). "A Structured Approach to Teaching Recursion Using Cargo-Bot". In: *Proceedings of the Tenth Annual Conference on International Computing Education Research*. ICER '14. Glasgow, Scotland, United Kingdom: Association for Computing Machinery, pp. 59–66. ISBN: 9781450327558. DOI: [10.1145/2632320.2632356](https://doi.org/10.1145/2632320.2632356). URL: <https://doi.org/10.1145/2632320.2632356>

1.2.1 Description

Professors at The University of Texas at Austin used an iPad game called "Cargo-Bot" to teach their approximately 200 undergraduate students recursion. The game involved students moving boxes from an original configuration into a goal configuration by using recursion. The game involves several puzzles and the students were also given lectures with examples from the game (Lee et al., 2014).

1.2.2 *Role*

The audience for this resource was approximately 200 undergraduate computer science students. They were learning recursion from the beginning. The professors were trying to use this game to help students understand recursion at a high level (the game does not involve code), which they hoped would translate to better recursive code (Lee et al., 2014).

1.2.3 *Educationally Strong Points*

The students who completed more than seven of the puzzles in Cargo-Bot had statistically significantly higher scores than students who did not use Cargo-Bot (Lee et al., 2014). Cargo-Bot did not involve code but rather the high-level concept of recursion. Other research also found that teaching students the high-level concept was useful. For example, professors at Cleveland State University used a live demonstration of the Towers of Hanoi and students self reported that this was either neutral or beneficial to their overall understanding of recursion (Benander and Benander, 2008). At Williams College, professors also found that students were receptive to structural and real-life examples like the rings of a bulls-eye for understanding recursion at a high level (Bruce, Danyluk, and Murtagh, 2005).

1.2.4 *Improving Educational Value*

Professors at Williams College found that structural and real-life recursion examples helped students learn recursion (Bruce, Danyluk, and Murtagh, 2005). However, they also paired these examples with real code and they explained recursion before they taught them other iterative techniques in code, such as using arrays (Bruce, Danyluk, and Murtagh, 2005). Other researchers found that combining visual examples (like drawings of fractals) alongside real code gave students multiple perspectives of the same problem and helped their understanding (Elenbogen and O'Kennon, 1988). At Carnegie-Mellon they had students write code which in turn made a visualization of the Tower of Hanoi problem run, which students self-reported helped their understanding (Dann, Cooper, and Pausch, 2001). Cargo-Bot could be taught alongside code and prior to learning arrays, both of which may make it more effective for novice programmers. In addition, when professors at Cleveland State University showed students the live Towers of Hanoi problem, they also had those students take notes about the

recursive calls and values (Benander and Benander, 2008). This also might be a useful activity for students so they can better connect the recursion concept to actual code.

1.3 Online Python Tutor: Embeddable Web-Based Program Visualization for CS Education

Philip J Guo (2013). “Online python tutor: embeddable web-based program visualization for cs education”. In: *Proceeding of the 44th ACM technical symposium on Computer science education*, pp. 579–584

1.3.1 Description

This project was first developed by a graduate student who had difficulty teaching computer science concepts, including recursion, through traditional PowerPoint slides or whiteboard diagrams (Guo, 2013). It was then expanded upon by a team at Google to help teachers around the world visualize Python code for their students easily for free. The website originally just visualized Python code and recursive call stacks but can now visualize other languages as well. Users who visit the site can input code and the Online Python Tutor will act as the interpreter and run the code line-by-line, updating values as it goes (Guo, 2013).

1.3.2 Role

This resource is attempting to replace the recursive call stack diagrams that teachers typically made either by using PowerPoint or by hand on a whiteboard. The audience was originally the author (a graduate student who was teaching a class) and his students but expanded and now the audience is novice programmers around the world. It is not opinionated and can be used for users learning recursion for the first time or as a refresher, or students learning a new language and attempting to visualize the problems in that language. The Online Python Tutor is also not just for recursion but for any code that the user wants to visualize, though the paper does specifically mention recursive call stacks as being one of the main use cases (Guo, 2013).

1.3.3 Educationally Strong Points

The authors of an e-book for computer science that also included visualizations found that students who used those visualizations outside of class time had higher test scores (Alvarado et al., 2012). While it is unclear if using those vi-

sualizations impacted their learning or if students who were already technically strong tended to do more work on their own time, either way the Online Python Tutor probably has similar outcomes on students. In addition, professors found that students who used SRec, a Java-based visualization of recursive calls, completed assignments faster with fewer errors (Velázquez-Iturbide and Pérez-Carrasco, 2016). This also shows that visualizations help students understand code better.

1.3.4 *Improving Educational Value*

Professors in Colombia found that students benefited from being able to submit their code as many times as needed and from receiving specific error codes to help guide them (Sánchez and Salazar, 2020). This could be incorporated into the Online Python Tutor by giving sample problems with tests and then allowing students to visualize the code they wrote line-by-line until the error is thrown. Similarly, other researchers created a tool that compared code to ten million potential fixes and then gave the students specific feedback on how to fix their code (Singh, Gulwani, and Solar-Lezama, 2013). This would also require the Online Python Tutor to have sample problems and answers, but it would help students who are having trouble walking through their code get more specific feedback so they can increase their knowledge quickly.

2 ROLE OF COLLABORATION IN LEARNING

2.1 Qualifier Question

Second, consider the role of collaboration in learning. Your A3 discussed the possibilities of a discussion board component. What value does collaboration offer to learning? When is it most effective? Discuss two peer-reviewed studies that show the benefits of collaboration in learning, taking care to note how the ideas presented could apply to an online learning tool.

2.2 Value of Collaboration

2.2.1 *Introduction*

Collaboration can be very beneficial to learning. Collaboration allows students to "exercise, verify, solidify, and improve their mental models through discussion and information sharing" (Leidner and Jarvenpaa, 1995). Students not only increase their own understanding but also the understanding of the other stu-

dents in the group; it is mutually beneficial (O'Neill, Scott, and Conboy, 2011).

2.2.2 A Delphi Study on Collaborative Learning in Distance Education: The Faculty Perspective

Researchers at NUI Galway investigated "the most important factors that influence the effective use of collaborative learning in distance education" by giving eighteen experts a survey (O'Neill, Scott, and Conboy, 2011). The experts were all educational leaders who used collaborative techniques in distance learning. The researchers found 28 initial factors from literature, and then asked their experts to use that as a stepping stone to indicate important ones and add any that were missed. The surveys were then evaluated and re-sent out in phases until they had a ranked list of the most important factors. They also asked experts to similarly rank technologies that assisted in collaboration. They found that according to these experts, the most important factors included activity design, teaching style, and accessibility, and the highest-ranked technologies were online forums and document tools (such as Google Docs) (O'Neill, Scott, and Conboy, 2011).

An interesting finding in this study was that teaching style was considered to have a high influence on participation in the online collaborative technologies (O'Neill, Scott, and Conboy, 2011). This implies that it is not enough for these technologies to be available to students, but that instructors must encourage students to use them in order to be effective.

2.2.3 Improving Student Learning of Conceptual Information: GSS Supported Collaborative Learning vs. Individual Constructive Learning

Researchers at Baylor University compared students who worked individually to students who worked collaboratively to see if there were measurable differences in their learning (Leidner and Fuller, 1997). In this study, professors used two sections of their undergraduate business class for comparison. Both groups were given case studies to work on, but they alternated so that sometimes each section would work in small groups, sometimes large groups, and sometimes individually. They used assignment scores and forms to evaluate each of these. They found that while students enjoyed working together more and perceived themselves as learning more in groups, students who worked individually had higher scores on the assignments (Leidner and Fuller, 1997).

This is an interesting study because the students who did not collaborate actually performed better. However, potentially a more interesting note is that students who collaborated on the assignment self reported being more interested in the subject and learning as a whole (Leidner and Fuller, 1997). This could mean that collaboration encourages more learning outside of class. Additionally, those students enjoyed the assignments more, which could have long-term effects (pursuing a thesis, for example). The long-term implications of this study were not explored, however the enjoyment of learning via collaboration is an important aspect to keeping students' interest.

2.2.4 Conclusion

These two studies show that collaboration is an important factor in learning. Students enjoy learning more when they collaborate and they increase the learning of those around them as well. In a distance learning format, this can include online forums or online document tools such as Google Docs. Collaborative learning can potentially be combined with individual learning to get the best of both worlds. If students are encouraged to participate and find that they gain new ideas from their peers via an online forum, this could be an invaluable tool for students in distance education during their individual study time. For example, if the online forum allowed students to share resources that helped them learn (videos, blogs, etc.), this could help them share information asynchronously so their individual study time is more effective. The study found that small groups versus large groups did not matter for learning, so the online discussion boards can be inclusive to the whole class (Leidner and Fuller, 1997). In conclusion, collaboration can make distance learning more effective and interesting for students.

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